SQL Joins – Hands-on Assignments

1. Natural Join

Question:
Write a query for the HR department to produce the addresses of all the departments. Use the EMP and DEPT tables. Show the **EMPNO, ENAME, SAL, DNAME, and LOC** in the output. Use a **NATURAL JOIN** to produce the results.

Solution:-

```
SELECT EMPNO, ENAME, SAL, DNAME, LOC

FROM EMP

NATURAL JOIN DEPT;
```

---

2. Equi Join

Question:
Write a query to display the **JOB, MGR, SAL, COMM, DNAME** of employees whose **JOB** is **SALESMAN**.

Solution:-

```
SELECT E.JOB, E.MGR, E.SAL, E.COMM, D.DNAME

FROM EMP E, DEPT D

WHERE E.DEPTNO = D.DEPTNO

AND E.JOB = 'SALESMAN';
```

---

3. Equi Join

Question:
Display the **ENAME, JOB, DEPTNO, DNAME** for all employees who work in **DALLAS**.

Solution:-

```
SELECT E.ENAME, E.JOB, E.DEPTNO, D.DNAME

FROM EMP E, DEPT D
```

```
WHERE E.DEPTNO = D.DEPTNO
AND D.LOC = 'DALLAS';
```

---

4. Self Join

Question:
Create a report to display employees' name and employee number along with their manager's name and manager number. Label the columns **Employee, Emp#, Manager, Mgr#** respectively.

Solution:-

```
SELECT
   E.ENAME AS Employee,
   E.EMPNO AS "Emp#",
   M.ENAME AS Manager,
   M.EMPNO AS "Mgr#"
FROM EMP E
JOIN EMP M
ON E.MGR = M.EMPNO;
```

---

5. Outer Join

Question:
Modify the previous query to display all employees including **KING**, who has no manager. Order the results by employee number.

Solution:-

```
SELECT
   E.ENAME AS Employee,
```

```
    E.EMPNO AS "Emp#",
    M.ENAME AS Manager,
    M.EMPNO AS "Mgr#"
FROM EMP E
LEFT OUTER JOIN EMP M
ON E.MGR = M.EMPNO
ORDER BY E.EMPNO;
```

---

6. Non-Equi Join

Question:
First show the structure of the **SALGRADE** table. Then display the **name, job, department name, salary, and grade** for all employees.

Solution:-

```
DESC SALGRADE;
SELECT
    E.ENAME,
    E.JOB,
    D.DNAME,
    E.SAL,
    S.GRADE
FROM EMP E
JOIN DEPT D
ON E.DEPTNO = D.DEPTNO
JOIN SALGRADE S
ON E.SAL BETWEEN S.LOSAL AND S.HISAL;
```

---

7. Outer Join

Question:
Display the **ENAME** and **DNAME** of all employees. Also display department names that do not have any employees working.

Solution:-

```
SELECT E.ENAME, D.DNAME
FROM EMP E
RIGHT OUTER JOIN DEPT D
ON E.DEPTNO = D.DEPTNO;
```

---

8. Self Join with Outer Join

Question:
Find the names and hire dates for all employees who were hired before their managers, along with their managers' names and hire dates.

Solution:-

```
SELECT
   E.ENAME AS Employee,
   E.HIREDATE AS Employee_HireDate,
   M.ENAME AS Manager,
   M.HIREDATE AS Manager_HireDate
FROM EMP E
LEFT OUTER JOIN EMP M
ON E.MGR = M.EMPNO
WHERE M.HIREDATE > E.HIREDATE;
```

---

9. USING Clause

Question:
Display the **EMPNO, ENAME, DNAME, LOC** of employees who are working as **CLERK**. Use the **USING** clause.

Solution:-

```
SELECT EMPNO, ENAME, DNAME, LOC

FROM EMP

JOIN DEPT

USING (DEPTNO)

WHERE JOB = 'CLERK';
```

---

10. ON Clause

Question:
Display the **ENAME, SAL, MGR, DNAME** of employees whose salary is more than **2000**. Use the **ON** clause.

Solution:-

```
SELECT E.ENAME, E.SAL, E.MGR, D.DNAME

FROM EMP E

JOIN DEPT D

ON E.DEPTNO = D.DEPTNO

WHERE E.SAL > 2000;
```

---

11. LEFT OUTER JOIN

Question:
Display the **EMPNO, ENAME, JOB, DEPTNO, DNAME, LOC** of employees. Use **LEFT OUTER JOIN**.

Solution:-

```
SELECT

  E.EMPNO,
```

```
  E.ENAME,
  E.JOB,
  E.DEPTNO,
  D.DNAME,
  D.LOC
FROM EMP E
LEFT OUTER JOIN DEPT D
ON E.DEPTNO = D.DEPTNO;
```

---

12. RIGHT OUTER JOIN

Question:
Display the **ENAME** and **DNAME** of employees. Use **RIGHT OUTER JOIN**.

Solution:-

```
SELECT E.ENAME, D.DNAME
FROM EMP E
RIGHT OUTER JOIN DEPT D
ON E.DEPTNO = D.DEPTNO;
```

---

13. FULL OUTER JOIN

Question:
Display the **EMPNO, DNAME, LOC** of employees. Use **FULL OUTER JOIN**.

Solution:-

```
SELECT
  E.EMPNO,
  D.DNAME,
  D.LOC
FROM EMP E
```

```
FULL OUTER JOIN DEPT D
ON E.DEPTNO = D.DEPTNO;
```